

M01 - Data Foundations + Essential Excel

15-lesson learner book | Beginner-first professional institute standard | B01-style teaching contract

Purpose

Understand data grain/quality and use essential Excel without damaging source data.

Course boundary

This is the independent Standalone Data Engineer course, not Four-Course C3. Complete the course in sequence; each new idea is still taught from first principles.

Option B institute identity | Charcoal #1F2937 | Teal #14B8A6 | Soft Blue #3B82F6

This book is your teacher

Use one lesson at a time. Do not read protected review before saving a genuine attempt. Every lesson follows the same beginner-first sequence even when the engineering idea becomes more technical.

Daily lesson loop

1) why it matters -> 2) what you learn -> 3) prerequisite -> 4) picture it -> 5) new words -> 6) useful visual/source -> 7) follow once -> 8) expected result -> 9) why it worked -> 10) mistakes -> 11) recovery -> 12) independent task -> 13) validate -> 14) teach it back -> 15) protected review + changed retry -> 16) evidence/mastery + exact next route.

Prerequisite check

Complete M00 file/folder safety and basic working-copy habits. You should be able to locate a file, preserve an original, and save evidence.

Attempt protection

A protected review is repair material, not a shortcut. Save the first attempt, open only the matching review if needed, record the repair, close review, then solve a changed retry from a clean state.

Contents / Index

Every formal lesson and its actual page number in this exact rebuilt book.

DE01-01 - What data is: tables, records, fields and data types	5
DE01-02 - Grain: what one row represents	7
DE01-03 - Keys, duplicates, NULLs and missing data	9
DE01-04 - Data quality dimensions and validation thinking	11
DE01-05 - Meet Excel: workbook, worksheet, cells and formulas	13
DE01-06 - Enter, save and structure tabular data safely	15
DE01-07 - Sort and filter without damaging data	17
DE01-08 - Basic formulas and essential functions	19
DE01-09 - Relative and absolute references	21
DE01-10 - Excel Tables and structured data habits	23
DE01-11 - Practical cleaning and lookup awareness	25
DE01-12 - PivotTable and Power Query awareness	27
DE01-13 - Chart choice and basic visual communication	29
DE01-14 - Reconciliation, sanity checks and source preservation	31
DE01-15 - Mini workflow: inspect -> calculate -> validate -> explain -> save evidence	33

Prerequisite, tool and outcome map

ASSUMED

Complete M00 file/folder safety and basic working-copy habits. You should be able to locate a file, preserve an original, and save evidence.

MODULE PURPOSE

Understand data grain/quality and use essential Excel without damaging source data.

EXIT EVIDENCE

Module practical + fresh retest + saved evidence; no critical unresolved failure.

Protected study route

Teaching -> assigned visual/source when useful -> follow once -> close worked material -> independent attempt -> validate -> save evidence -> teach it back. If weak: matching protected review -> targeted repair -> close review -> changed fresh retry -> revalidate.

DE01-01 - What data is: tables, records, fields and data types

1. Why it matters

Data tools cannot be used safely until you know what one row and each field mean.

2. What you are learning

Understand what a table, row, column, cell and data type are before you touch formulas or SQL.

3. Prerequisite / what you already know

Complete M00 file/folder safety and basic working-copy habits. You should be able to locate a file, preserve an original, and save evidence.

4. Picture it in your head

Picture a table as a stack of index cards: one card is a record/row, each labeled box on the card is a field/column, and the data type says what kind of value belongs in that box.

5. New words before we use them

Word Plain-English meaning Table A rectangular collection of related data. Row / record One observation or event at the table's current level of detail. Column / field One attribute stored consistently across rows. Cell One value at the intersection of a row and column. Data type The kind of value: text, whole number, decimal, date/time, true/false, etc.

6. Useful video / source

Source: Luke Barousse — Data Engineer Bootcamp (2026)

Assigned scope: 4:16:18 Data Modeling & JOINS + 6:57:49 Data Types

How to use it: Use the mapped chapters as visual reinforcement for tables/fields/types. The Teaching Book supplies the beginner definitions and independent proof.

[OPEN VIDEO / SOURCE - CLICK HERE \(youtube.com\)](https://www.youtube.com/watch?v=ol9_NnC9-cc)

URL: https://www.youtube.com/watch?v=ol9_NnC9-cc

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

- Open the workbook sheet Raw_Sales.
- Look only at the header row first: OrderID, OrderDate, CustomerID, CustomerName, Region, Product, Qty, UnitPrice, PaymentMethod.
- Pick row 2. Read it left to right as one order record.
- Classify Qty and UnitPrice as numbers, OrderDate as a date-like value, and Region/Product/PaymentMethod as text categories.
- G01 - Map table grain and field types

8. Expected result / what you should see

You should be able to point at any value and say both what column it belongs to and what business thing the entire row represents.

9. Why it worked

The result is trustworthy because structure is treated as a contract rather than inferred casually; the validation checks fields/types/constraints against the declared shape. Your independent check is: For What data is: tables, records, fields and data types: record the actual before/after state and verify the lesson requirement with an independent control. The proof is the preserved table/formula/control, not how the sheet looks on screen.

10. Common beginner mistakes

- A value looking like a date does not guarantee Excel stored it as a true date.
- Formatting is appearance; data type is how the value behaves.
- Do not assume every number should be added. OrderID contains digits but is an identifier, not a quantity.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Picture a table as a stack of index cards: one card is a record/row, each labeled box on the card is a field/column, and the data type says what kind of value belongs in that box. 3. Run the earliest independent control: For What data is: tables, records, fields and data types: record the actual before/after state and verify the lesson requirement with an independent control. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

P01 - Grain and candidate key In Raw_Sales, choose OrderID, Qty, UnitPrice and PaymentMethod. Write whether each is identifier, numeric measure or text category, and explain why.

13. Validate your result

For What data is: tables, records, fields and data types: record the actual before/after state and verify the lesson requirement with an independent control. Challenge this expected outcome directly: You should be able to point at any value and say both what column it belongs to and what business thing the entire row represents.

14. Teach it back in your own words

Without notes, explain What data is: tables, records, fields and data types in your own words. Start with the purpose: Understand what a table, row, column, cell and data type are before you touch formulas or SQL. Then name one realistic failure mode and explain how you would prove the result using this check: For What data is: tables, records, fields and data types: record the actual before/after state and verify the lesson requirement with an independent control.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for What data is: tables, records, fields and data types with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE01-02.

DE01-02 - Grain: what one row represents

1. Why it matters

Grain controls counting, duplicates, joins and summaries; wrong grain creates believable but wrong answers.

2. What you are learning

Learn grain - the sentence that says exactly what one row represents.

3. Prerequisite / what you already know

Open Raw_Sales + Support_Tickets. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

Think of grain as camera zoom: "one row per order" and "one row per order item" are different zoom levels. A calculation is trustworthy only when you know which level the row represents.

5. New words before we use them

Word Plain-English meaning Grain The business meaning of one row. Order header Usually one row per order. Order line Usually one row per product line inside an order. Aggregation Combining many lower-level rows into fewer summary rows.

6. Useful video / source

Internal teaching route

Internal teaching route. This lesson is fully taught in this book and its bundled practice. Use the mental model and worked demonstration here; no filler external video is required for mastery.

7. Follow with me once

- Look at Raw_Sales. Ask: if OrderID O1001 appears once, what event does that row describe?
- Write: One row represents one order record for one customer/product purchase in this practice data.
- Now imagine an order-line table where one order buys three products. That same OrderID could legitimately appear three times because the grain is one order line, not one order.
- G01 - Map table grain and field types

8. Expected result / what you should see

Before counting, joining or summarizing, you can say "one row represents ." in plain English.

9. Why it worked

This works because learn grain - the sentence that says exactly what one row represents is made explicit before the result is accepted. Your independent check is: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. The proof is the preserved table/formula/control, not how the sheet looks on screen.

10. Common beginner mistakes

- Saying "grain = customer" just because CustomerID exists.
- Counting rows as orders without proving the table is one row per order.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Think of grain as camera zoom: "one row per order" and "one row per order item" are different zoom levels. 3. Run the earliest independent control: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

P01 - Grain and candidate key + P02 - Legitimate repeats versus duplicates For Support_Tickets, write the grain sentence. Then explain why CustomerID may repeat even if TicketID should not.

13. Validate your result

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. Before counting, joining or summarizing, you can say "one row represents ." in plain English.

14. Teach it back in your own words

Without notes, explain Grain: what one row represents in your own words. Start with the purpose: Learn grain - the sentence that says exactly what one row represents. Then name one realistic failure mode and explain how you would prove the result using this check: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for Grain: what one row represents with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE01-03.

DE01-03 - Keys, duplicates, NULLs and missing data

1. Why it matters

A repeated value is only a duplicate relative to the table grain and key. Missingness also needs a business rule, not automatic deletion.

2. What you are learning

Tell legitimate repeats from suspicious duplicates and understand missing values.

3. Prerequisite / what you already know

Open Raw_Sales + Support_Tickets. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

Imagine every real entity carrying an ID badge. A key should identify the intended record, duplicates are two badges claiming the same identity, and NULL means the value is unknown/not supplied rather than zero.

5. New words before we use them

Word Plain-English meaning
Key A column or combination of columns used to identify the row at its intended grain.
Duplicate A repeated row/key that should be unique at that grain.
NULL / blank A missing or unknown value.
Zero A known numeric value equal to nothing - not the same as missing.

6. Useful video / source

Source: Luke Barousse — Data Engineer Bootcamp (2026)

Assigned scope: 4:16:18 Data Modeling & JOINS + 10:16:15 Date/Text/NULL Functions

How to use it: Watch only the relevant chapter blocks. Teaching Book adds duplicate detection and the course validation task.

[OPEN VIDEO / SOURCE - CLICK HERE \(youtube.com\)](https://www.youtube.com/watch?v=ol9_NnC9-cc)

URL: https://www.youtube.com/watch?v=ol9_NnC9-cc

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

- In Raw_Sales, find the two O1019 rows. Compare every column. They are identical, so this is suspicious at one-order-record grain.
- Find CustomerID C003 appearing on more than one order. That is not automatically wrong because one customer can place multiple orders.
- Find the blank CustomerID and blank PaymentMethod. Treat these as missing-information questions, not as zero.
- G02 - Key and duplicate inspection + G03 - Missingness and validity scan

8. Expected result / what you should see

You do not delete a repeat until you can explain why it violates the key/grain.

9. Why it worked

This works because tell legitimate repeats from suspicious duplicates and understand missing values is made explicit before the result is accepted. Your independent check is: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. The proof is the preserved table/formula/control, not how the sheet looks on screen.

10. Common beginner mistakes

- Using "Remove Duplicates" before deciding the key.
- Replacing blanks with 0 just to make them disappear.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Imagine every real entity carrying an ID badge. 3. Run the earliest independent control: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

P02 - Legitimate repeats versus duplicates In Raw_Sales, write: candidate key, one legitimate repeat, one suspicious duplicate, and two missing-value issues.

13. Validate your result

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. You do not delete a repeat until you can explain why it violates the key/grain.

14. Teach it back in your own words

Without notes, explain Keys, duplicates, NULLs and missing data in your own words. Start with the purpose: Tell legitimate repeats from suspicious duplicates and understand missing values. Then name one realistic failure mode and explain how you would prove the result using this check: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for Keys, duplicates, NULLs and missing data with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE01-04.

DE01-04 - Data quality dimensions and validation thinking

1. Why it matters

Quality checks turn vague trust into explicit tests that can fail visibly before bad data reaches reporting.

2. What you are learning

Turn "data quality" from a vague phrase into specific checks.

3. Prerequisite / what you already know

Open Raw_Sales. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

Picture a quality checkpoint at the exit of a pipeline: completeness, validity, uniqueness, consistency, timeliness, and reconciliation are separate inspection lights; one green light cannot prove the others.

5. New words before we use them

Word Plain-English meaning
Completeness Are required values present?
Validity Do values obey allowed rules/ranges?
Uniqueness Are values that should be unique actually unique?
Consistency Do the same concepts use consistent labels/formats?
Timeliness Is the data recent enough for the decision?

6. Useful video / source

Source: Statistics Canada — Data Quality in Six Dimensions

Assigned scope: Full 8:42 beginner video

How to use it: Watch the short official video, then use the Teaching Book to apply quality dimensions to engineering-style validation.

OPEN VIDEO / SOURCE - CLICK HERE ([statcan.gc.ca](https://www.statcan.gc.ca/en/wtc/data-literacy/catalogue/892000062020001))

URL: <https://www.statcan.gc.ca/en/wtc/data-literacy/catalogue/892000062020001>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

- Completeness: check CustomerID/PaymentMethod blanks.
- Validity: Qty should normally be positive; Region should belong to the approved set.
- Uniqueness: OrderID should be unique if grain is one order.
- Consistency: East and east should not silently become different business regions.
- Timeliness: an old dataset may be correct but too stale for today's operational decision.
- G03 - Missingness and validity scan

8. Expected result / what you should see

Every quality claim becomes a concrete test: column + rule + what failure looks like.

9. Why it worked

This works because turn "data quality" from a vague phrase into specific checks is made explicit before the result is accepted. Your independent check is: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. The proof is the preserved table/formula/control, not how the sheet looks on screen.

10. Common beginner mistakes

- Writing "check data quality" without a test.
- Confusing uniqueness with "no repeated values anywhere".
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Picture a quality checkpoint at the exit of a pipeline: completeness, validity, uniqueness, consistency, timeliness, and reconciliation are separate inspection lights; one green light cannot prove the others. 3. Run the earliest independent control: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

P03 - Design five data-quality checks Using Raw_Sales, write one specific check for each of the five dimensions.

13. Validate your result

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. Every quality claim becomes a concrete test: column + rule + what failure looks like.

14. Teach it back in your own words

Without notes, explain Data quality dimensions and validation thinking in your own words. Start with the purpose: Turn "data quality" from a vague phrase into specific checks. Then name one realistic failure mode and explain how you would prove the result using this check: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for Data quality dimensions and validation thinking with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE01-05.

DE01-05 - Meet Excel: workbook, worksheet, cells and formulas

1. Why it matters

Excel instructions depend on precise workbook, worksheet and cell language; navigation mistakes later become formula mistakes.

2. What you are learning

Understand the Excel screen: workbook, worksheet, cell, address, formula bar and formula.

3. Prerequisite / what you already know

Open START_HERE + workbook working copy. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

A workbook is like a notebook file; worksheets are the tabs/pages inside it, cells are coordinates on a page, and a formula is an instruction stored behind the displayed result.

5. New words before we use them

Word Plain-English meaning
Workbook The .xlsx file.
Worksheet A tab inside the workbook.
Cell address Column letter + row number, such as D2.
Formula bar The area that shows the actual value/formula stored in the active cell.
Formula An instruction beginning with = that Excel calculates.

6. Useful video / source

Source: Chandoo — Excel Tutorial for Beginners

Assigned scope: 1:15–4:09

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[OPEN VIDEO / SOURCE - CLICK HERE \(youtube.com\)](https://www.youtube.com/watch?v=F7aPazuS8QY)

URL: <https://www.youtube.com/watch?v=F7aPazuS8QY>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

- Open M01_EXCEL_PRACTICE_WORKBOOK_TEACHING_RC2.xlsx.
- Click the Formula_Practice tab.
- Click cell D2. Notice the Name Box/selected cell address says D2.
- Type =B2*C2 and press Enter.
- Click D2 again. The cell shows a result, while the formula bar shows =B2*C2.
- G04 - Workbook navigation proof

8. Expected result / what you should see

You understand the difference between what the cell displays and the formula stored behind it.

9. Why it worked

The workbook/worksheet/cell model works because each value and formula lives at a known address and Excel recalculates from those explicit references. Prove it by changing one intended input, confirming the dependent formula changes, and confirming an unrelated cell does not.

10. Common beginner mistakes

- Typing 1700 manually instead of =B2*C2.
- Typing B2*C2 without the leading =.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: A workbook is like a notebook file; worksheets are the tabs/pages inside it, cells are coordinates on a page, and a formula is an instruction stored behind the displayed result. 3. Run the earliest independent control: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

Checkpoint 2 - navigation/formula setup proof Click three different cells and say their addresses aloud. Then write one simple addition formula in an unused cell and undo it.

13. Validate your result

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. You understand the difference between what the cell displays and the formula stored behind it.

14. Teach it back in your own words

Without notes, explain Meet Excel: workbook, worksheet, cells and formulas in your own words. Start with the purpose: Understand the Excel screen: workbook, worksheet, cell, address, formula bar and formula. Then name one realistic failure mode and explain how you would prove the result using this check: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for Meet Excel: workbook, worksheet, cells and formulas with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE01-06.

DE01-06 - Enter, save and structure tabular data safely

1. Why it matters

Preserving the source and using a working copy makes mistakes reversible and keeps evidence reproducible.

2. What you are learning

Enter and save tabular data without destroying the original source.

3. Prerequisite / what you already know

Open Raw_Sales working copy. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

Think of tabular data as a rectangular form: one header row defines the fields, each following row is one record, and blank decorative rows/merged cells break the machine-readable rectangle.

5. New words before we use them

Word Plain-English meaning Raw data Original received data before your transformations. Working copy A separate file you are allowed to edit. Header row The row containing column names. Tabular structure One header row, one observation per row, one attribute per column.

6. Useful video / source

Source: Chandoo — Excel Tutorial for Beginners

Assigned scope: 4:09–8:21

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[OPEN VIDEO / SOURCE - CLICK HERE \(youtube.com\)](https://www.youtube.com/watch?v=F7aPazuS8QY)

URL: <https://www.youtube.com/watch?v=F7aPazuS8QY>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

- Open the workbook. Use File > Save As and create a working copy such as M01_working.xlsx.
- Keep one header row. Avoid merged cells inside data tables.
- Do not insert decorative blank rows in the middle of the dataset.
- Save frequently with Ctrl+S.
- G04 - Workbook navigation proof + safe working-copy discipline

8. Expected result / what you should see

The original workbook remains unchanged, and your working copy contains your experiments.

9. Why it worked

This works because enter and save tabular data without destroying the original source is made explicit before the result is accepted. Your independent check is: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. The proof is the preserved table/formula/control, not how the sheet looks on screen.

10. Common beginner mistakes

- Editing the only copy of the raw file.
- Using two header rows or merged cells as decoration inside source data.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Think of tabular data as a rectangular form: one header row defines the fields, each following row is one record, and blank decorative rows/merged cells break the machine-readable rectangle. 3. Run the earliest independent control: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

Checkpoint 2 - safe working-copy and structure proof Create a working copy, rename it clearly, and write the full filename in your evidence notes.

13. Validate your result

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. The original workbook remains unchanged, and your working copy contains your experiments.

14. Teach it back in your own words

Without notes, explain Enter, save and structure tabular data safely in your own words. Start with the purpose: Enter and save tabular data without destroying the original source. Then name one realistic failure mode and explain how you would prove the result using this check: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for Enter, save and structure tabular data safely with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE01-07.

DE01-07 - Sort and filter without damaging data

1. Why it matters

Filtering should change only what you see; sorting should change order without breaking row integrity.

2. What you are learning

Use Sort and Filter as viewing/ordering tools without deleting or misaligning data.

3. Prerequisite / what you already know

Open Raw_Sales working copy. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

Sorting is rearranging cards on the table; filtering is temporarily hiding cards that do not match a rule. Neither should silently delete or overwrite source records.

5. New words before we use them

Word Plain-English meaning Filter Temporarily shows rows matching conditions; hidden rows remain in the table. Sort Changes row order based on one or more columns. Whole-row integrity Values from the same record must move together.

6. Useful video / source

Source: Chandoo — Excel Tutorial for Beginners

Assigned scope: 11:37–15:46

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[OPEN VIDEO / SOURCE - CLICK HERE \(youtube.com\)](https://www.youtube.com/watch?v=F7aPazuS8QY)

URL: <https://www.youtube.com/watch?v=F7aPazuS8QY>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

- In a working copy of Raw_Sales, click inside the data.
- Choose Data > Filter.
- Filter Region to East/east and note the visible rows.
- Clear the filter and confirm the full set returns.
- For sorting, click inside the table and sort the entire data range by UnitPrice - do not select only UnitPrice cells in isolation.
- G05 - Safe sort/filter

8. Expected result / what you should see

Filtering never deletes the hidden rows; sorting keeps OrderID, customer, product, Qty and price aligned on the same record.

9. Why it worked

The spreadsheet result is trustworthy because the source structure/formula logic is preserved and an independent cell/row/total check proves the displayed result came from the intended calculation. Your independent check is: For Sort and filter without damaging data: record the actual before/after state and verify the lesson requirement with an independent control. The proof is the preserved table/formula/control, not how the sheet looks on screen.

10. Common beginner mistakes

- Deleting hidden rows because they are not visible.
- Sorting one column by itself and scrambling row relationships.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Sorting is rearranging cards on the table; filtering is temporarily hiding cards that do not match a rule. 3. Run the earliest independent control: For Sort and filter without damaging data: record the actual before/after state and verify the lesson requirement with an independent control. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

P04 - Filter and prove no deletion Filter one PaymentMethod, note the visible count, clear it, and confirm the original row count returns.

13. Validate your result

For Sort and filter without damaging data: record the actual before/after state and verify the lesson requirement with an independent control. Challenge this expected outcome directly: Filtering never deletes the hidden rows; sorting keeps OrderID, customer, product, Qty and price aligned on the same record.

14. Teach it back in your own words

Without notes, explain Sort and filter without damaging data in your own words. Start with the purpose: Use Sort and Filter as viewing/ordering tools without deleting or misaligning data. Then name one realistic failure mode and explain how you would prove the result using this check: For Sort and filter without damaging data: record the actual before/after state and verify the lesson requirement with an independent control.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for Sort and filter without damaging data with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE01-08.

DE01-08 - Basic formulas and essential functions

1. Why it matters

Formulas make calculations repeatable and auditable; manual arithmetic is used as an independent control, not as the production method.

2. What you are learning

Create basic formulas and use SUM, AVERAGE and COUNT/COUNTA.

3. Prerequisite / what you already know

Open Formula_Practice. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

A formula is a recipe attached to a cell: it takes referenced ingredients, performs an operation, and recalculates when the ingredients change.

5. New words before we use them

Word Plain-English meaning
Cell reference A pointer to a cell such as B2.
SUM Adds numeric values.
AVERAGE Calculates arithmetic mean.
COUNT Counts numeric cells.
COUNTA Counts nonblank cells.

6. Useful video / source

Source: Chandoo — Excel Tutorial for Beginners

Assigned scope: 15:46–22:39

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

OPEN VIDEO / SOURCE - CLICK HERE ([youtube.com](https://www.youtube.com/watch?v=F7aPazuS8QY))

URL: <https://www.youtube.com/watch?v=F7aPazuS8QY>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

- Open Formula_Practice. In D2 type =B2*C2.
- Fill/copy D2 down through the product rows.
- Manually check Keyboard: $2 \times 850 = 1700$.
- Use =SUM(D2:D7) for total Sales.
- Use =AVERAGE(D2:D7) for average Sales.
- Use COUNT on a numeric column or COUNTA on a populated text/ID column depending on what you intend to count.
- G06 - Basic row formulas

8. Expected result / what you should see

Changing a Qty or UnitPrice automatically changes the calculated Sales; formulas appear in the formula bar.

9. Why it worked

The program is trustworthy because input/state changes are explicit and the output is checked against a known condition rather than accepting “the script ran” as correctness. Your independent check is: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. The proof is the preserved table/formula/control, not how the sheet looks on screen.

10. Common beginner mistakes

- Typing answers instead of formulas.
- Using COUNT on text IDs and thinking zero means there are no records.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: A formula is a recipe attached to a cell: it takes referenced ingredients, performs an operation, and recalculates when the ingredients change. 3. Run the earliest independent control: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

P05 - Revenue and Margin formulas Complete Formula_Practice and manually validate two rows before trusting the total.

13. Validate your result

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. Changing a Qty or UnitPrice automatically changes the calculated Sales; formulas appear in the formula bar.

14. Teach it back in your own words

Without notes, explain Basic formulas and essential functions in your own words. Start with the purpose: Create basic formulas and use SUM, AVERAGE and COUNT/COUNTA. Then name one realistic failure mode and explain how you would prove the result using this check: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for Basic formulas and essential functions with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE01-09.

DE01-09 - Relative and absolute references

1. Why it matters

Relative references should move with rows; constants such as a tax/rate cell must stay fixed when copied.

2. What you are learning

Understand why A2 moves when copied but \$H\$2 stays fixed.

3. Prerequisite / what you already know

Open Reference_Practice. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

Picture copying a formula as moving a map instruction. A relative reference moves with you; an absolute reference is pinned to a fixed coordinate with \$ signs.

5. New words before we use them

Word Plain-English meaning Relative reference A2; changes relative to the formula location when copied. Absolute reference \$H\$2; row and column are locked. Mixed reference \$H2 or H\$2; only one part is locked.

6. Useful video / source

Source: Chandoo — Useful Excel concepts

Assigned scope: 12:35–20:52

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[OPEN VIDEO / SOURCE - CLICK HERE \(youtube.com\)](https://www.youtube.com/watch?v=VT479YDPB0I)

URL: <https://www.youtube.com/watch?v=VT479YDPB0I>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

- Open Reference_Practice. Calculate Base Sales in D2 using Qty x UnitPrice.
- Tax Rate is stored once in H2. In E2 use a formula such as =D2*(1+\$H\$2).
- Copy E2 down. Click E3, E4, E5. D2 changes to D3/D4/D5, but \$H\$2 stays exactly \$H\$2.
- Temporarily change H2. Every taxed result should update because all formulas point to the same rate cell. Undo if needed.
- G07 - Relative vs absolute reference

8. Expected result / what you should see

One setting cell controls every row without repeating the percentage in each formula.

9. Why it worked

This works because understand why A2 moves when copied but \$H\$2 stays fixed is made explicit before the result is accepted. Your independent check is: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. The proof is the preserved table/formula/control, not how the sheet looks on screen.

10. Common beginner mistakes

- Using H2 without dollar signs and watching it drift to H3/H4.
- Typing 18% directly in every row formula.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Picture copying a formula as moving a map instruction. 3. Run the earliest independent control: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

P06 - One absolute rate cell Complete Independent_Rate using the single rate in H2 and explain what each dollar sign protects.

13. Validate your result

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. One setting cell controls every row without repeating the percentage in each formula.

14. Teach it back in your own words

Without notes, explain Relative and absolute references in your own words. Start with the purpose: Understand why A2 moves when copied but \$H\$2 stays fixed. Then name one realistic failure mode and explain how you would prove the result using this check: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for Relative and absolute references with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE01-10.

DE01-10 - Excel Tables and structured data habits

1. Why it matters

Excel Tables create explicit rectangular structure, stable headers and predictable expansion/filter behavior.

2. What you are learning

Know the difference between a normal range and an actual Excel Table.

3. Prerequisite / what you already know

Open clean working range. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

An Excel Table is a fenced data region that knows its headers and expands with new rows, so formulas and references can follow the structure instead of fragile cell ranges.

5. New words before we use them

Word Plain-English meaning Excel Table A structured Excel object created with Insert > Table or Ctrl+T. Header Named field at the top of a Table column. Structured behavior Filters, automatic expansion and formula propagation tied to the Table object.

6. Useful video / source

Source: Chandoo — Useful Excel concepts

Assigned scope: 2:22–8:45

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[OPEN VIDEO / SOURCE - CLICK HERE \(youtube.com\)](https://www.youtube.com/watch?v=VT479YDPB0I)

URL: <https://www.youtube.com/watch?v=VT479YDPB0I>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

- Work on a clean copy/range. Click inside it.
- Choose Insert > Table (or Ctrl+T).
- Confirm the full range and tick My table has headers.
- Notice filter arrows appear. Add one new row directly below the Table and see the Table expand.
- G08 - Excel Table behaviour

8. Expected result / what you should see

The object expands with new rows and keeps explicit headers/filters.

9. Why it worked

An Excel Table works because the rectangular structure, headers and structured references travel with the rows instead of depending on fragile fixed ranges. Prove it by adding one valid row, checking table expansion/formula propagation, and reconciling the new row count and total.

10. Common beginner mistakes

- Assuming coloured formatting means the data is an Excel Table.
- Creating a Table from an incomplete range that omits columns.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: An Excel Table is a fenced data region that knows its headers and expands with new rows, so formulas and references can follow the structure instead of fragile cell ranges. 3. Run the earliest independent control: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

P07 - Prove Excel Table behaviour Convert a small clean range to a Table, add a row, then describe one behavior that proves it is a Table.

13. Validate your result

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. The object expands with new rows and keeps explicit headers/filters.

14. Teach it back in your own words

Without notes, explain Excel Tables and structured data habits in your own words. Start with the purpose: Know the difference between a normal range and an actual Excel Table. Then name one realistic failure mode and explain how you would prove the result using this check: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for Excel Tables and structured data habits with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE01-11.

DE01-11 - Practical cleaning and lookup awareness

1. Why it matters

Cleaning should standardize representation without inventing business meaning; lookups connect tables by a deliberate key.

2. What you are learning

Understand basic cleaning and how XLOOKUP connects related tables using an ID.

3. Prerequisite / what you already know

Open Raw_Sales / PowerQuery_Source + Customer_Lookup. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

A lookup is like using an ID in an index to fetch a related attribute. The key must be trustworthy before the returned value is trustworthy.

5. New words before we use them

Word Plain-English meaning
Cleaning Making values consistent/usable without changing their business meaning.
Lookup value The ID you are trying to find. Lookup array Where Excel searches for that ID. Return array The column from which Excel returns the answer. Not found The explicit result when no match exists.

6. Useful video / source

Source: Chandoo — XLOOKUP for Beginners

Assigned scope: Whole short video

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[OPEN VIDEO / SOURCE - CLICK HERE \(youtube.com\)](https://www.youtube.com/watch?v=dIBuWllegRg)

URL: <https://www.youtube.com/watch?v=dIBuWllegRg>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

- Notice Region values East, east and West with trailing space in Raw_Sales. These are consistency problems.
- Open Customer_Lookup. CustomerID is the linking field; Tier is the value you want to bring back.
- In Independent_Lookup, use XLOOKUP with CustomerID as the lookup value, Customer_Lookup!CustomerID as search range, and Tier as return range.
- Handle C999 explicitly with a not-found message rather than silently assuming a tier.
- G09 - Cleaning awareness + lookup follow-along in learner book

8. Expected result / what you should see

Known IDs return the same Tier shown in Customer_Lookup; C999 is clearly flagged missing.

9. Why it worked

The cleaning-and-lookup workflow works because keys are standardized before matching and unmatched values remain visible instead of being silently forced into a result. Prove it with pre/post row counts, an unmatched-key check, and spot-checks of matched values against the source.

10. Common beginner mistakes

- Matching by CustomerName when CustomerID is the intended key.
- Treating unmatched IDs as blank and forgetting to investigate them.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: A lookup is like using an ID in an index to fetch a related attribute. 3. Run the earliest independent control: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

P08 - Lookup Tier by CustomerID Complete Independent_Lookup, spot-check two known customers, and explain the four pieces of XLOOKUP in plain English.

13. Validate your result

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. Known IDs return the same Tier shown in Customer_Lookup; C999 is clearly flagged missing.

14. Teach it back in your own words

Without notes, explain Practical cleaning and lookup awareness in your own words. Start with the purpose: Understand basic cleaning and how XLOOKUP connects related tables using an ID. Then name one realistic failure mode and explain how you would prove the result using this check: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for Practical cleaning and lookup awareness with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE01-12.

DE01-12 - PivotTable and Power Query awareness

1. Why it matters

PivotTables summarize quickly; Power Query records repeatable import/cleaning logic. At this stage you need a safe mental model, not advanced BI mastery.

2. What you are learning

Understand what PivotTables and Power Query are for without trying to master them yet.

3. Prerequisite / what you already know

Open Pivot_Source + PowerQuery_Source. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

Picture a PivotTable as a movable summary board and Power Query as a recorded cleaning recipe. One summarizes a prepared table; the other repeats transformation steps before analysis.

5. New words before we use them

Word Plain-English meaning PivotTable Interactive summary built from source data. Rows area Categories listed vertically. Values area Measures summarized, e.g., Sum of Sales. Power Query Excel's repeatable import/transform tool. Refresh Run the saved import/transform logic again on updated data.

6. Useful video / source

Source: Chandoo — PivotTable / Power Query awareness

Assigned scope: 31:35–38:35

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[OPEN VIDEO / SOURCE - CLICK HERE \(youtube.com\)](https://www.youtube.com/watch?v=VT479YDPB0I)

URL: <https://www.youtube.com/watch?v=VT479YDPB0I>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

- Pivot: click Pivot_Source > Insert > PivotTable > New Worksheet. Put Region in Rows and Sales in Values.
- Check that Values says Sum of Sales, not Count of Sales.
- Power Query awareness: inspect PowerQuery_Source. Find leading/trailing spaces, inconsistent case, mixed date formats and Amount stored as text with comma.
- The point is not to hand-fix every cell. A Power Query workflow can record trim/clean/type/standardization steps and refresh them later.
- Pivot follow-along in learner book + P09 after guided attempt

8. Expected result / what you should see

You can answer which Region has the highest sales and describe Power Query as repeatable transformation rather than "another worksheet".

9. Why it worked

The calculation is trustworthy because the output grain is declared before aggregation and the grouped result is reconciled to an independent detail-level control. Your independent check is: For PivotTable and Power Query awareness: Recompute one grouped total from detail rows and reconcile the sum of group totals to the known overall control; confirm the result has one row per declared group. The proof is the preserved table/formula/control, not how the sheet looks on screen.

10. Common beginner mistakes

- Building a fancy dashboard before understanding the summary.
- Using Count of Sales by accident.
- Manually cleaning source values one-by-one and calling that a repeatable pipeline.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Picture a PivotTable as a movable summary board and Power Query as a recorded cleaning recipe. 3. Run the earliest independent control: For PivotTable and Power Query awareness: Recompute one grouped total from detail rows and reconcile the sum of group totals to the known overall control; confirm the result has one row per declared group. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

P09 - Pivot summary and reconciliation Create the simple PivotTable, reconcile its grand total to source SUM, then write 2-3 sentences explaining Power Query.

13. Validate your result

For PivotTable and Power Query awareness: Recompute one grouped total from detail rows and reconcile the sum of group totals to the known overall control; confirm the result has one row per declared group.

14. Teach it back in your own words

Without notes, explain PivotTable and Power Query awareness in your own words. Start with the purpose: Understand what PivotTables and Power Query are for without trying to master them yet. Then name one realistic failure mode and explain how you would prove the result using this check: For PivotTable and Power Query awareness: Recompute one grouped total from detail rows and reconcile the sum of group totals to the known overall control; confirm the result has one row per declared group.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for PivotTable and Power Query awareness with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE01-13.

DE01-13 - Chart choice and basic visual communication

1. Why it matters

A chart should answer the business question clearly; decorative visuals can hide the comparison instead of explaining it.

2. What you are learning

Choose a simple chart that matches the question and avoid decorative charts that hide the message.

3. Prerequisite / what you already know

Open Pivot_Source / summary range. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

Charts are lenses: bars compare categories, lines show ordered change over time, and scatter plots show relationships. The wrong lens can make correct numbers tell a confusing story.

5. New words before we use them

Word Plain-English meaning Bar/column chart Good for comparing categories. Line chart Good for trends over ordered time. Pie chart Limited use; difficult when many categories/slices exist. Axis Scale along horizontal/vertical direction. Title Should state what is being compared, not just "Chart 1".

6. Useful video / source

Source: Chandoo — What is a Chart and How to Make Charts in Excel (video)

Assigned scope: Embedded beginner video: chart purpose, create/format/change chart type; use the page's "select correct chart" follow-up only as needed

How to use it: Watch only enough to understand chart choice and basic communication. Then use the Teaching Book task with your own data; do not turn this into a dashboard course.

OPEN VIDEO / SOURCE - CLICK HERE (chandoo.org)

URL: <https://chandoo.org/wp/excel-tutorial/making-charts/>

Do this after watching

Close the external source. Return to the bundled lesson and perform the exact Follow with me once steps from your own keyboard. Save evidence from your course data/tool, not a screenshot of the video.

7. Follow with me once

- Use the regional summary from your PivotTable.
- For comparing East/West/North/South, choose a bar or column chart.
- Give it a clear title such as "Sales by Region"
-
- Check that the largest bar visually matches the largest numeric total.
- Simple chart follow-along + Checkpoint 4 visual task

8. Expected result / what you should see

A reader can answer the business question quickly without decoding unnecessary colours or 3-D effects.

9. Why it worked

The chart works because the visual form matches the business question while the underlying values, categories and scale remain traceable to the source table. Prove it by reconciling plotted values to source cells and confirming the chart communicates the intended comparison or trend without hiding an important scale or category.

10. Common beginner mistakes

- Using a line chart for unordered categories.
- Adding 3-D effects that distort comparisons.
- Making the title vague.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Charts are lenses: bars compare categories, lines show ordered change over time, and scatter plots show relationships. 3. Run the earliest independent control: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

Checkpoint 4 - create a simple comparison chart Create one simple regional comparison chart and write one sentence explaining why that chart type fits the question.

13. Validate your result

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. A reader can answer the business question quickly without decoding unnecessary colours or 3-D effects.

14. Teach it back in your own words

Without notes, explain Chart choice and basic visual communication in your own words. Start with the purpose: Choose a simple chart that matches the question and avoid decorative charts that hide the message. Then name one realistic failure mode and explain how you would prove the result using this check: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for Chart choice and basic visual communication with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE01-14.

DE01-14 - Reconciliation, sanity checks and source preservation

1. Why it matters

Plausible output is not proof. Reconciliation and source preservation make a result independently checkable and recoverable.

2. What you are learning

Learn to distrust even plausible-looking results until you reconcile them to an independent check.

3. Prerequisite / what you already know

Open completed working copy + source data. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

Reconciliation is like closing a cash register: independently compare row counts, keys, and important totals to the source before trusting a cleaned or summarized result.

5. New words before we use them

Word Plain-English meaning Reconciliation Comparing the result to an independent total/count/control. Source preservation Keeping raw evidence unchanged. Sanity check A quick reasonableness check for impossible or surprising values. Control total A known count/amount used to verify processing.

6. Useful video / source

Internal teaching route

Internal teaching route. This lesson is fully taught in this book and its bundled practice. Use the mental model and worked demonstration here; no filler external video is required for mastery.

7. Follow with me once

- Before transformations, record source row count and any important total.
- After filtering/cleaning/calculating, recompute the total another way.
- If Pivot grand total differs from SUM of source Sales, stop and find the earliest step where rows/values diverged.
- Do not "fix the final number" manually; repair the logic.
- Reconciliation follow-along + Checkpoint 4

8. Expected result / what you should see

You can show both the result and one independent check that supports it.

9. Why it worked

The workflow is trustworthy because the received input is identified and preserved before transformation, so every downstream result can be traced back to unchanged source evidence. Your independent check is: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. The proof is the preserved table/formula/control, not how the sheet looks on screen.

10. Common beginner mistakes

- Trusting a pretty PivotTable because it looks professional.
- Changing raw data to force agreement.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Reconciliation is like closing a cash register: independently compare row counts, keys, and important totals to the source before trusting a cleaned or summarized result. 3. Run the earliest independent control: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

P09 + Checkpoint 4 - reconcile to independent total For Pivot_Source, compare Pivot grand total with =SUM(E2:E19). Record both and explain whether they reconcile.

13. Validate your result

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. You can show both the result and one independent check that supports it.

14. Teach it back in your own words

Without notes, explain Reconciliation, sanity checks and source preservation in your own words. Start with the purpose: Learn to distrust even plausible-looking results until you reconcile them to an independent check. Then name one realistic failure mode and explain how you would prove the result using this check: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for Reconciliation, sanity checks and source preservation with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE01-15.

DE01-15 - Mini workflow: inspect -> calculate -> validate -> explain -> save evidence

1. Why it matters

A professional workflow is a chain: inspect, calculate, validate, explain and save evidence. Skipping any link makes handoff fragile.

2. What you are learning

Combine the module into a repeatable beginner workflow: inspect -> calculate -> validate -> explain -> save evidence.

3. Prerequisite / what you already know

Open one completed M01 working file. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

Use a lab notebook mental model: first attempt, observed result, repair, and changed retry are separate entries. If you erase the failed attempt, you lose the evidence of learning.

5. New words before we use them

Word Plain-English meaning
Inspect Understand columns, grain, keys and quality before calculating.
Calculate Apply formulas/summaries that answer the question.
Validate Use an independent check.
Explain State result and reasoning in plain language.
Evidence Saved workbook, note, screenshot, formula or output proving what you actually did.

6. Useful video / source

Internal teaching route

Internal teaching route. This lesson is fully taught in this book and its bundled practice. Use the mental model and worked demonstration here; no filler external video is required for mastery.

7. Follow with me once

- Read the business question first.
- State the table grain and candidate key.
- Preserve raw source and work on a copy.
- Perform only the transformations/calculations needed.
- Validate with row counts, control totals or spot checks.
- Explain the answer, one limitation and one possible failure mode.
- Save the workbook and note where the evidence lives.
- Module workflow rehearsal before practical

8. Expected result / what you should see

Someone else can reproduce and audit your work instead of trusting an unexplained final number.

9. Why it worked

The reasoning separates correctness from performance: first prove the result, then inspect the planner/execution evidence to see whether the engine can avoid unnecessary work. Your independent check is: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. The proof is the preserved table/formula/control, not how the sheet looks on screen.

10. Common beginner mistakes

- Starting with a chart/formula before understanding the data.
- Saving only screenshots and losing the actual workbook/formulas.
- Reporting "done" with no validation.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck - recovery ladder

1. Return to the lesson prerequisite/starting state; do not stack fixes on an unknown state. 2. Rebuild the mental picture: Use a lab notebook mental model: first attempt, observed result, repair, and changed retry are separate entries. 3. Run the earliest independent control: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. 4. Repair only the first failed step, rerun that same control, and save the failed attempt before opening protected Review.

12. Now you do a small version alone

P10 - Mini analyst handoff + Module Practical Write a 5-8 sentence handoff for one M01 practice task: source, grain, what you changed, result, validation, limitation.

13. Validate your result

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. Someone else can reproduce and audit your work instead of trusting an unexplained final number.

14. Teach it back in your own words

Without notes, explain Mini workflow: inspect -> calculate -> validate -> explain -> save evidence in your own words. Start with the purpose: Combine the module into a repeatable beginner workflow: inspect -> calculate -> validate -> explain -> save evidence. Then name one realistic failure mode and explain how you would prove the result using this check: Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

15. Protected review + changed fresh retry

Save the genuine independent attempt first. If it is weak, open only the matching protected Review, identify the exact gap, record the repair, then close Review. Do a changed retry for Mini workflow: inspect -> calculate -> validate -> explain -> save evidence with a different input/identifier/condition and validate it independently. Work completed while Review is open is repair evidence, not mastery evidence.

16. Evidence / mastery + exact next route

Mastery evidence: Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: module practical / Gate route after the mastery checklist.